

HyperGranular-RAG: Benefits and Boundaries of Structured Evidence Completion in Multi-Hop Question Answering

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Abstract

Multi-hop retrieval-augmented generation requires deciding which evidence to select and how to arrange it within a limited evidence budget. A relevant sentence may need a complementary bridge, yet introducing that bridge can displace useful context. We examine these decisions through HyperGranular-RAG, a structured evidence-completion layer that organizes candidate sentences by embedding geometry, selects seed-relative facet hyperedges, and inserts a bounded number of candidates after a protected dense prefix. With Qwen2.5-1.5B-Instruct, compact completion improves answer F1 over MiniLM Dense by 0.01478, 0.01140, and 0.01357 in three closed-candidate HotpotQA and MuSiQue evaluations; the recorded 95% paired bootstrap intervals exclude zero. The specified compact facet selector also exceeds a centroid-only alternative. In contrast, the original compact system falls below BGE, and neither tested BGE extension establishes an increment over that baseline. Generator transfer remains unresolved. Protected placement outperforms front insertion with identical visible evidence in the cross-space study, distinguishing an ordering benefit from an established gain over BGE. Supporting-evidence turnover and analysis of group-level versus selected-unit coverage help interpret this distinction. The results support completion for the tested compact backbone while leaving strong-backbone complementarity and the superiority of granular-ball organization over simpler selection rules unresolved.

Keywords: retrieval-augmented generation; multi-hop question answering; evidence selection; granular representations; uncertainty.

1 Introduction

Retrieval-augmented question answering depends on two distinct capabilities: finding relevant information and supplying a generator with a usable combination of that information. A strong relevance ranking addresses the first capability, but does not necessarily satisfy the second. For a question requiring an entity bridge, several highly relevant sentences about the first entity may be less useful than one additional sentence connecting it to the second. Dense retrieval (Karpukhin et al., 2020) provides a natural backbone for this process, while multi-hop benchmarks expose the need to assemble related facts (Yang et al., 2018; Trivedi et al., 2022).

A finite evidence-list budget makes completion a replacement decision. If a system retains at most twenty evidence units, moving an additional candidate into the context can remove another unit. The new item may increase annotated support coverage without improving the answer; it may also duplicate retained evidence or move information to a position the generator uses poorly. Position-sensitive context use has been documented in language models (Liu et al., 2024). Consequently, the number of retrieved supporting facts, their placement, and the answer produced from them are related but non-interchangeable outcomes.

HyperGranular-RAG studies a bounded intervention on this ranking. Candidate sentence embeddings are organized into local granular balls. Query terms define facets over these balls, and seed-relative facet hyperedges identify candidate regions that differ from the initial seeds. A placement policy retains the

first ten dense entries and supplies at most four candidates before filling the remaining positions from the dense order. The method is an evidence-completion layer: it preserves a retrieval backbone and does not replace corpus search with a learned graph model.

Two linked decisions organize our study: which evidence should enter the final list, and how should that evidence be arranged? Full-versus-Dense evaluates the completion pipeline, while Full-versus-NoFacet compares its selector with the specified centroid-only rule. The cross-space Protected-versus-Unprotected contrast fixes visible evidence and tests its ordering. Comparisons with BGE and an additional generator examine how far the resulting benefits extend. Evidence-turnover and geometric analyses then delimit what these comparisons explain about selection and arrangement.

The empirical answer is conditional. Across three Qwen evaluations, the compact implementation improves over MiniLM Dense. In the joint component study, the facet-conditioned selector exceeds the specified centroid-only control. Nevertheless, the original compact system is worse than BGE. Adding compact candidates to BGE and rebuilding the structure in BGE space both produce intervals crossing zero. The positive compact effects therefore establish a capability within that system, while the BGE results limit claims about general retrieval improvement.

The implementations answer related but distinct questions. Compact completion uses MiniLM for both ranking and organization; the cross-space sidecar combines BGE ranking with MiniLM candidates; the native variant reconstructs geometry and scoring in BGE space. The latter changes more than the encoder. We specify these differences before interpreting their results together.

The article contributes an explicitly defined evidence-completion layer, an empirical account of its benefits and boundaries, and an analysis separating evidence selection from arrangement. Repeated compact gains and the specified facet comparison provide the positive evidence. Strong-backbone and generator comparisons establish its current scope, while the fixed-visible-evidence contrast identifies the tested placement-policy effect. The extended analysis connects these results to supporting-evidence turnover and the limits of group-level coverage as an explanation of answer quality.

2 Positioning within Retrieval and Evidence Selection

2.1 From Relevance Ranking to Connected Retrieval

Dense passage retrieval learns query and passage encoders for relevance matching (Karpukhin et al., 2020). Multi-hop retrieval introduces dependencies between retrieval steps. MDR conditions a subsequent retrieval on preceding evidence (Xiong et al., 2021), whereas IRCOT alternates retrieval with generated reasoning (Trivedi et al., 2023). Those methods change how a system reaches evidence. Our closed-candidate intervention instead changes which already available units enter a bounded context.

This distinction determines the scope of evaluation. A method can improve selection from a supplied candidate set without improving first-stage corpus recall. Conversely, a corpus retriever can reach the correct document without selecting the sentence the generator needs. The present study addresses the second type of problem conditional on candidate availability. It therefore does not claim to solve open-domain reachability.

2.2 Granular-Ball Organization and Hypergraph Learning

Granular-ball computing supplies a representation of groups at different scales (Xia et al., 2019); it is more specific than granular computing in general. SAGHL uses graph topology to construct hierarchical hyperedges and couples them to learned representations (Ni et al., 2026). The MGHRL preprint partitions graph nodes using shortest-path assignments and combines multiple hypergraph-network columns (Zhao et al., 2026). Granular-ball divergence and fuzzy rough hypergraphs also support feature selection (Li et al., 2026). These precedents cover the general combination of granular balls and hypergraphs. We do not claim to originate it.

Family	Structural object and operation	Relation to this study
SAGHL / MGHRL	Graph-node groups and learned hypergraph representations	Construction precedents; not QA baselines
Fuzzy rough feature selection	Granular-ball discrimination and feature relations	Coverage/selection precedent; different utility target
HGRAG (Wang et al.)	Entity–passage incidence and query-conditioned diffusion	Closest QA neighbor; different corpus, extraction and reader
HyperGraphRAG / Hyper-RAG	Extracted multi-entity relations, retrieval and text fusion	Factual hyperedges, unlike lexical group-selection relations
HyperGranular-RAG (ours)	Sentence-embedding groups, static facets, bounded placement	Evaluated local completion; no established advantage over the above systems

Table 1: Conceptual comparisons, not reproduced-system results. Names and structural objects are separated from performance claims. The MGHRL source is a preprint; the SAGHL and feature-selection comparison is limited to verified publisher descriptions.

Our evaluated implementation instead partitions sentence embeddings and uses the resulting groups for query-conditioned, inference-time selection. Its cosine geometry, static seed-relative scores and answer-quality endpoint differ from graph-topological learning or feature discrimination. These differences define an application and evaluation question; they do not by themselves establish a superior construction. In particular, shortest-path metric arguments cannot be transferred to one-minus-cosine dissimilarity, and classification improvements do not imply QA evidence utility.

2.3 Hypergraph and Cross-Granularity Retrieval

RAPTOR constructs a hierarchy through recursive abstraction (Sarathi et al., 2024). HippoRAG uses knowledge-graph structure and propagation (Gutiérrez et al., 2024). The closest task-level neighbor, HGRAG (Wang et al., AAAI 2026), treats extracted entities as nodes and passages as hyperedges (Wang et al., 2026). Query-dependent diffusion combines entity and passage similarity, followed by semantic mixing and structural context selection. Its pooled passage corpus, large encoder and reader, and variable passage budget differ from our per-question sentence candidates. We use the full name *HyperGranular-RAG* for our method to distinguish the two systems.

HyperGraphRAG retrieves LLM-extracted n-ary facts and fuses them with raw chunks (Luo et al., 2025). Hyper-RAG extracts low- and high-order relations, retrieves related entities and correlations, and evaluates domain-specific generation (Feng et al., 2026). These methods already establish hypergraphs as a RAG design choice. Their reported gains are not measurements under our candidates, generator, scoring function or budget. Table 1 summarizes the representational distinctions.

In HyperGranular-RAG, a granular ball is a local group of sentence embeddings with a center and radius, inspired by granular representations (Xia et al., 2019). A facet hyperedge relates seed balls to an expansion ball through query-term coverage. This relation is not an asserted real-world fact and does not result from an entity-relation extraction model. The structure is query-local, computed from the available candidate set, and used to rank a small expansion. The historical study contains no budget-aligned direct comparison with these external systems. Incidence storage is not the relevant difference: a bipartite graph with explicit hyperedge identities can preserve membership losslessly, as also described by Luo et al. (2025). Removing those identities or altering selection is a different intervention from changing storage format. Likewise, a richer representation does not guarantee a better answer when extraction noise, evidence displacement and a finite context intervene.

2.4 Coverage and Diversity as Alternative Explanations

A structured selector may improve a ranking simply by avoiding redundant evidence. MMR is a classical relevance–redundancy rule (Carbonell and Goldstein, 1998). Explicit coverage and diversity objectives can offer another route to evidence selection (Lin and Bilmes, 2011). These alternatives matter because a useful grouping heuristic does not by itself establish that high-order relations caused the benefit.

We use the term *facet-conditioned* to describe the implemented rule, not to settle that causal question. Its compact NoFacet comparison replaces the selector with centroid-only ball expansion. It does not hold every possible coverage or diversity principle constant, and it does not compare the method to a sentence-level coverage optimizer. The distinction between a pipeline comparison and a pure component intervention is maintained throughout the analysis.

2.5 Context Use and Empirical Uncertainty

Long-context evaluations show that evidence location can affect model performance (Liu et al., 2024). This motivates protecting an initial retrieval prefix, but does not prove that ten entries is an optimal protection budget. Likewise, statistical testing guidance emphasizes the relationship between a statistic, a null hypothesis, and the dependence structure of the sample (Dror et al., 2018). We report historical estimates with their recorded intervals and describe the evaluator and inferential caveats explicitly, rather than strengthening a result through terminology.

3 Evidence Selection and Placement

3.1 Candidate, Ranking, and Prompt Spaces

Let q denote a question and U_q the finite set of sentence units supplied by its benchmark context. A unit includes a document title, sentence text, and a stable identity. The encoder receives the title and sentence, and produces a normalized embedding $\hat{e}_u$. With query embedding $\hat{e}_q$, the dense relevance score is

$$s(q, u) = \hat{e}_q^\top \hat{e}_u. \quad (1)$$

Units are ordered by decreasing score and increasing identity. The baseline list D_q contains the first $K_q = \min(20, |U_q|)$ unique units. The completed list R_q has the same length.

Three evidence objects must be distinguished. The *candidate insertion set* contains units accepted by an expansion policy. The *final ranking* is the ordered list after placement, deduplication, and truncation to K_q . The *prompt-visible evidence* is the text actually retained by the token-limited prompt. Equality at one level does not automatically imply equality at the next. In particular, a fixed number of sentence units is not a fixed number of tokens.

3.2 Compact Granular-Ball Organization

The compact method uses MiniLM embeddings for both relevance and grouping. It begins with the whole candidate set as one root. For a ball B ,

$$c_B = \text{norm} \left(\sum_{u \in B} \hat{e}_u \right), \quad d(u, B) = 1 - \hat{e}_u^\top c_B, \quad r_B = \max_{u \in B} d(u, B). \quad (2)$$

Compactness is $C(B) = 1 - |B|^{-1} \sum_{u \in B} d(u, B)$. The compact split eligibility rule is

$$\text{depth}(B) < 6 \quad \wedge \quad |B| \geq 4 \quad \wedge \quad (|B| > 3 \vee r_B > 0.78). \quad (3)$$

Because $|B| \geq 4$ implies $|B| > 3$, the radius disjunction is inactive. The first split seed is farthest from the center; the second has minimum cosine similarity to the first. Assignment uses the more similar seed and the frozen tie rule. A split is accepted only when each child has at least two members. Otherwise the unsplit group becomes a leaf, possibly with more than three members.

Thus the compact partition adapts to size and geometry, but is not independently radius-triggered. The compactness analysis in Section 4 explains why geometrically more compact children are not, on their own, validated semantic evidence units.

3.3 Facet Hyperedges and Static Selection

The fixed tokenizer extracts ASCII-alphanumeric content terms and removes the recorded stop words. Let T_q denote query terms and $T_B = T_q \cap \bigcup_{u \in B} T(u)$ the facets of ball B , using its title and sentence text. The two balls with highest query-centroid similarity are the seeds. Their union of facets, T_{seed} , defines the reference for expansion.

For a non-seed ball, write $n_B = |T_B \setminus T_{\text{seed}}|$, $t_B = |T_B|$, $m = |T_q|$, $b_B = \hat{e}_q^\top c_B$, $a_B = \max_{S \in \text{seeds}} c_B^\top c_S$, and $\rho_B = |T_B \cap T_{\text{seed}}|/|T_B|$ when T_B is nonempty. The candidate score is

$$h_B = .45 \frac{n_B}{\max(m, 1)} + .20 \frac{t_B}{\max(m, 1)} + .20b_B + .15(1 - a_B) - .20\rho_B - .05|B|/8. \quad (4)$$

Eligibility requires $n_B \geq 1$, $|B| \leq 8$, $b_B \geq .12$ or $a_B \geq .05$, $|B|/n_B \leq 6$, $\rho_B \leq .90$, and $h_B \geq .10$. Eligible balls are ordered by decreasing h_B and stable edge identity; at most two distinct expansion balls are selected. Their units retain dense-score and identity order. The compact unit-level facet bonus is zero.

A facet hyperedge is the relation between the seed balls and an eligible expansion ball contributing seed-relative new facets. It identifies a region of candidates; it does not assert that every unit in that region supplies the missing fact. All eligibility and score calculations occur once. Updating an internal covered-term set after accepting a ball does not recompute the remaining scores. This static rule is different from true marginal-coverage greedy selection.

3.4 Protected Bounded Insertion

Let P_q be the first $\min(10, K_q)$ entries of D_q . The compact method traverses the expansion order, skips duplicates in P_q , and accepts at most four units with dense score at least 0.1957079917192459. The floor is a development-calibrated 25th percentile, transferred unchanged. It is not a percentile recomputed on each evaluation set. The policy concatenates the protected prefix, accepted units, and remaining dense entries, applies stable deduplication, and truncates to K_q .

An accepted unit can already be in dense positions 11–20. Such an event is a promotion within the baseline set; it differs from introducing a new unit outside that set. Both consume placement capacity. This distinction matters when interpreting insertion counts as evidence expansion. Candidate counts cannot be substituted for counts of genuinely new supporting facts. Figure 1 separates selection, placement, and prompting.

3.5 Cross-Space Sidecar

The sidecar holds BGE’s dense ranking as the reference while the frozen MiniLM structure proposes eligible units outside BGE Top-20. It introduces no learned score fusion. Protected placement inserts its accepted candidates after the BGE prefix; unprotected placement puts the same candidates first. Both then fill from BGE and enforce the same effective list length.

The sidecar asks whether a compact structure contains useful complementary evidence that BGE does not already expose. It is more targeted than comparing the original MiniLM pipeline against BGE,

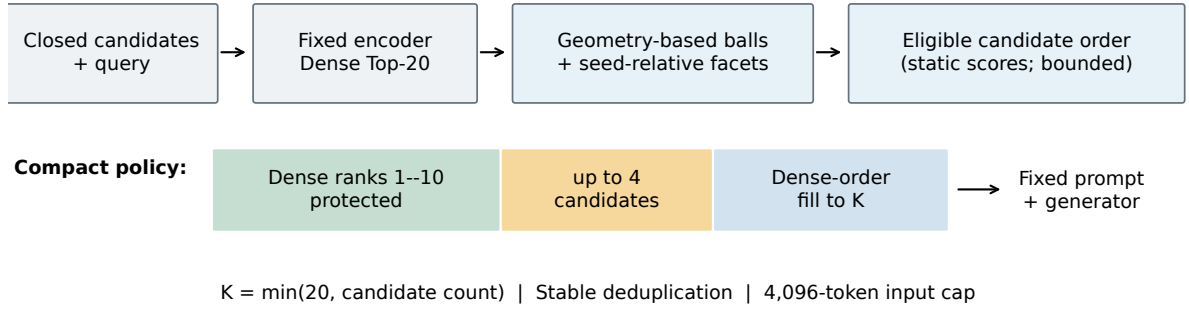


Figure 1: Compact completion. The root includes the full candidate pool. Static facet selection precedes placement; final list length and the generator’s token cap are separate constraints.

because the primary ranking now comes from BGE in both arms. It remains a cross-space construction: the candidate source and its score geometry are not native to BGE.

3.6 BGE-Native Reconstruction

The native implementation changes that geometry. A separate development family crosses leaf target $\{4, 6\}$, insertion budget $\{2, 4\}$, unit-score percentile floor $\{.50, .75\}$, and broad/strict facet gates, giving 16 configurations. The selected C10 uses leaf target six, budget four, floor .50, and broad gates. The confirmation comparison fixes that choice.

Native splits use depth limit eight, minimum child fraction .20, and a weighted-child compactness improvement greater than 10^{-10} . Seed selection uses deterministic center-relative two-seed geometry and identity-based ties. Within a query, unit relevance, ball relevance, radius, compactness, and density are converted to empirical percentiles, where the percentile of a value is the fraction of values less than or equal to it. Density is group size divided by radius with the recorded small denominator floor.

Two seed balls are selected by relevance percentile, density percentile, and identity. Broad gates require ball-relevance percentile at least .35 and density percentile at least .25; strict gates use .50 for both. Eligible expansion balls must have a seed-relative new facet and are ranked by

$$h_B^N = .55p_B + .25p_{\rho,B} + .15p_{C,B} + .05 \min(n_B, 3)/3. \quad (5)$$

At most four balls supply units. A candidate must be outside BGE Top-20 and meet the configured unit percentile floor. Its score is

$$h_u^N = .45p_u + .25p_B + .15p_{\rho,B} + .10p_{C,B} + .05 \min(n_B, 3)/3. \quad (6)$$

Native NoFacet removes the new-facet gate and uses $.50p_u + .30p_B + .10p_{\rho,B} + .10p_{C,B}$ for units. It therefore changes several coefficients as well as facet eligibility. Its result is a pipeline contrast, not a coefficient-preserving pure ablation. The native study cannot be read as changing only the encoder in the compact algorithm.

4 Analytical Properties and Their Limits

4.1 Compactness Is a Resultant-Length Statistic

The following properties concern the mathematical representation. They are not claims about new empirical performance.

Proposition 1. For nonempty B with unit embeddings x_i , and nonzero $s_B = \sum_{i \in B} x_i$, the historical mean-cosine compactness satisfies

$$C(B) = \frac{1}{|B|} \sum_{i \in B} x_i^\top \frac{s_B}{\|s_B\|} = \frac{\|s_B\|}{|B|} \in [0, 1]. \quad (7)$$

The scalar has the continuous extension $C(B) = 0$ at $s_B = 0$.

Proof. Linearity of the inner product gives $\sum_i x_i^\top s_B / \|s_B\| = s_B^\top s_B / \|s_B\| = \|s_B\|$. The upper bound follows from $\|\sum_i x_i\| \leq \sum_i \|x_i\| = |B|$. Continuity follows from continuity of the norm. $\square$

The scalar extension does not determine a normalized center at zero. The compact implementation’s zero-vector convention and the native implementation’s fallback direction can consequently have different assignment behavior even though the exact zero-sum scalar is the same.

Proposition 2. For every partition $B = \bigsqcup_{j=1}^m B_j$ into nonempty groups,

$$\sum_j \frac{|B_j|}{|B|} C(B_j) \geq C(B). \quad (8)$$

Proof. Writing $s_j = \sum_{i \in B_j} x_i$, the left side is $|B|^{-1} \sum_j \|s_j\|$, while the right side is $|B|^{-1} \|\sum_j s_j\|$. The triangle inequality gives the claim. $\square$

Equality holds when the nonzero group sums share a direction. Otherwise a strict increase can arise without any semantic alignment. For four orthogonal unit vectors, parent compactness is $1/2$; every partition into two pairs has weighted compactness $1/\sqrt{2}$. A positive compactness change is therefore not evidence that a split recovered a reasoning relation. The native balance and depth rules constrain a procedure, but do not change this logical limitation.

4.2 Scalar Stability Does Not Guarantee Partition Stability

For corresponding perturbed vectors x'_i , the reverse triangle inequality yields

$$|C'(B) - C(B)| \leq \frac{\|\sum_i (x'_i - x_i)\|}{|B|} \leq \frac{1}{|B|} \sum_i \|x'_i - x_i\|. \quad (9)$$

This is a stability bound for the scalar with membership fixed. It does not bound changes in the normalized center near a zero sum, nor does it keep discrete assignments fixed. For example, $x_1 = (1, 0)$ and $x_2 = (-1, \epsilon)/\sqrt{1 + \epsilon^2}$ have a sum whose direction approaches the positive vertical axis as $\epsilon \downarrow 0$. Replacing ϵ with $-\epsilon$ makes the input perturbation arbitrarily small while reversing that limiting direction. A stability claim for a full partition would need additional separation assumptions.

4.3 Seed-Relative Novelty Is Not a Dynamic Marginal

Suppose seeds cover $\{\alpha\}$ and three candidate groups cover $\{\beta\}$, $\{\beta\}$, and $\{\gamma\}$. A static score order can select the first two. Both were new relative to the seeds, but the second adds no facet after the first is selected. A dynamic coverage rule would reconsider the remaining candidates at that point. This is a structural difference between algorithms, not an estimate of how often the event occurs in the historical corpus.

Likewise, let a selected group contain two sentences with facets $\{\alpha\}$ and $\{\beta\}$, while the placement rule emits only the first. The group union is $\{\alpha, \beta\}$ but selected-unit coverage is $\{\alpha\}$. We call their difference *phantom coverage*: a facet credited to the group but absent from the selected units. Prompt truncation can further reduce visible content. None of the historical aggregate results measures the prevalence or answer effect of this particular discrepancy.

4.4 A Coverage Guarantee Applies Only to Its Own Objective

To make the distinction concrete, consider a fixed set of units, fixed observable facets $T(u)$, and nonnegative weights w_t . Define

$$F(S) = \sum_t w_t \mathbf{1} \left[t \in \bigcup_{u \in S} T(u) \right]. \quad (10)$$

For a fixed protected set P , let $f_P(A) = F(P \cup A) - F(P)$. The residual objective is normalized, monotone, and submodular: as A grows, the set of new facets available from any additional unit can only shrink. This kind of explicitly specified coverage objective is consistent with the objective-based approach discussed by [Lin and Bilmes \(2011\)](#).

If at most b additional units are allowed and greedy selection recomputes the true marginal each step, then for an optimal additional set O ,

$$f_P(O) - f_P(A_i) \leq \sum_{u \in O \setminus A_i} [f_P(A_i \cup \{u\}) - f_P(A_i)] \leq b[f_P(A_{i+1}) - f_P(A_i)]. \quad (11)$$

Iterating gives $f_P(A_b) \geq [1 - (1 - 1/b)^b]f_P(O) \geq (1 - 1/e)f_P(O)$ for $b \geq 1$. The $b = 0$ case is trivial, and selecting all remaining units is optimal if fewer than b are available.

This is the standard greedy coverage bound, specialized to an explicitly defined surrogate under a cardinality budget and a fixed prefix. The historical static score in Equation 4 does not implement that algorithm. Variable token costs, negative penalties, changing facets, and generated answer F1 are outside the statement. In particular, a coverage guarantee cannot be used as an answer-quality guarantee.

5 Empirical Design

5.1 Data Boundaries and Research Questions

HotpotQA provides multi-hop questions with sentence-level supporting facts ([Yang et al., 2018](#)). MuSiQue constructs connected questions from single-hop components, with an answerable subset and a separate setting including unanswerable instances ([Trivedi et al., 2022](#)). We use the answerable training release, not the unanswerable extension. Both benchmarks provide the closed candidate contexts used in this study.

Table 2 separates the evaluations. The initial compact studies test one frozen policy on two datasets. The joint component set tests the same compact pipeline against additional baselines and component alternatives. The sidecar and native confirmation sets test different interventions around BGE. Their manifests record query-identity separation from earlier evaluation sets; the native confirmation also excludes its development sample. Generator transfer deliberately reuses the initial rankings and questions. Thus it is not an independent replication of candidate selection.

The selection comparisons evaluate complete pipelines and their specified facet selectors. The cross-space placement comparison instead holds visible content fixed and changes its order. BGE and generator comparisons test the scope of the resulting answer-quality effects. Post-decision evidence-turnover analyses address interpretation, not additional confirmation hypotheses.

5.2 Baselines and Component Comparisons

The compact baseline uses `sentence-transformers/all-MiniLM-L6-v2`. The strong baseline is `BAAI/bge-large-en-v1.5`, with a model revision recorded in the implementation manifest. The component evaluation also contains BM25 and a fixed Dense-BM25 hybrid. These comparisons hold the candidate boundary and generator fixed within an evaluation.

Compact NoFacet preserves granular-ball geometry but selects up to two eligible expansion balls by centroid relevance. NoProtection alters placement. The sidecar and native studies each contain BGE,

Evaluation	HotpotQA	MuSiQue	Scientific role
Initial compact	1,000	3,000	Two dataset-specific evaluations of frozen compact completion
Joint component	1,000	1,500	Seven-arm compact, lexical, hybrid, strong-dense, and component comparison
Cross-space sidecar	1,000	1,500	Four-arm comparison around BGE using compact candidates
Native development	500	750	Selection from a fixed 16-configuration family
Native confirmation	1,000	1,500	Evaluation of the selected configuration and fixed comparators
Generator transfer	1,000	3,000	Initial rankings reused with Gemma mobile-QAT

Table 2: Data boundaries. Counts are questions, not retrieval units. All come from training releases held out within the project, rather than official hidden benchmark tests. Development and confirmation are never pooled.

Protected, NoFacet, and Unprotected arms. Native NoFacet changes both facet eligibility and scoring coefficients, as defined above. These are the implemented controls, not interchangeable labels for a universal facet ablation.

A flat-unit granular-ball control was not completed because a unique matched mapping of ball seeds, split rules, and group budgets was not specified. That omission cannot establish either the necessity or the uselessness of granular balls. Random grouping, matched-count ordinary clustering, MMR, and sentence-level coverage were also not formal arms in these historical answer-quality evaluations.

5.3 Generation and Development Choices

Qwen2.5-1.5B-Instruct FP16 is the main generator. All main comparisons use a fixed short-answer evidence prompt, a 4,096-token input cap, greedy decoding, batch size one, and a maximum of 32 newly generated tokens. The instruction asks for the shortest answer using only the supplied evidence and UNKNOWN if it is insufficient. Outputs are stripped of leading and trailing whitespace. The recorded hardware is an RTX 4060 Laptop with 8GB VRAM.

An earlier 200-question generator-selection development study selected this Qwen deployment. The additional Gemma 4 E2B configuration runs in the official mobile-QAT format. The comparison is conditioned on hardware, prompt, cap, and deployable numerical format. It is not evidence that Qwen’s architecture is universally stronger or more efficient.

Native C10 was selected using a separate development evaluation, whose equal-weight answer-F1 difference was $+0.003143$. Its configuration was then fixed for confirmation. The development value is reported as selection context only; it is not averaged with the confirmation effect and does not override it. The pre-specified native search ended after confirmation.

5.4 Answer Scoring and Support Granularity

For each question, normalized answer exact match and token-overlap F1 measure the generated answer. Multiple reference aliases are maximized where the official dataset scorer specifies them. Mean answer F1 is the primary endpoint; EM provides a stricter exact-answer view.

We use the datasets’ pinned official answer functions. HotpotQA assigns zero F1 to unequal normalized answers involving yes/no/noanswer and to answers without token overlap. MuSiQue instead returns one when both normalized token lists are empty and maximizes EM and F1 independently over reference aliases; it does not apply HotpotQA’s categorical exception. HotpotQA retains its single reference answer.

A retrospective reconciliation applied these functions to unchanged predictions and bound reference answers. All confirmation and generator-transfer scores are unchanged. A common correction on one native development question changes absolute development means but no method difference or the reported C10 selection context; configuration selection was not rerun. Appendix A gives the scoring lineage, source versions, and record-level accounting. This additional scoring check is not new independent confirmation or an official leaderboard evaluation.

For supporting evidence, let G_q be the dataset’s annotated support identities and $V_q(R)$ those covered by ranking R . Then

$$\text{CR@20}(q) = \mathbf{1}[G_q \subseteq V_q(R_q)], \quad \text{ER@20}(q) = \frac{|G_q \cap V_q(R_q)|}{|G_q|}. \quad (12)$$

HotpotQA support identities are sentence-level. MuSiQue supplies supporting paragraphs, and coverage follows that annotation granularity. It is not MuSiQue sentence-level or complete-chain verification. These retrieval metrics concern ranked evidence before the generator’s use of the context.

5.5 Paired Estimation and Historical Inference

For methods a and b , question i , and dataset d , write $\delta_{di} = F1(a, di) - F1(b, di)$. A dataset-specific effect is $\hat{\Delta}_d = n_d^{-1} \sum_i \delta_{di}$. Joint studies target

$$\hat{\Delta}_{\text{eq}} = \frac{1}{2} \hat{\Delta}_{\text{HotpotQA}} + \frac{1}{2} \hat{\Delta}_{\text{MuSiQue}}. \quad (13)$$

This weights datasets equally rather than weighting all questions equally. The recorded procedure uses 10,000 paired query bootstrap samples; joint comparisons resample within each dataset and combine the two means. We independently reconstruct the original percentile intervals and reuse the same resampling indices for legacy and canonical scores. All confirmation intervals are unchanged. Evaluation boundaries are not pooled, and neither the bootstrap design nor the original hypothesis tests are replaced.

These intervals describe a query-resampling model conditional on the selected pipeline. They do not integrate uncertainty over configuration selection, pretraining, or alternate model initialization. Query-disjoint sets can still share source documents, and document-level dependence is not captured by the query bootstrap. Sample sizes reflect the resources allocated to the study, not a formal prospective power guarantee.

Some historical decision files label the nonpositive fraction of uncentered bootstrap differences as a one-sided p-value, followed by Holm adjustment for a specified family. A tail fraction computed around the observed empirical distribution does not automatically supply a null-calibrated p-value. Following the distinction between effect estimation and hypothesis testing (Dror et al., 2018), we retain those files as historical decisions but do not present their adjusted values as a newly validated familywise test. The manuscript reports effect estimates and recorded intervals, and describes whether those intervals exclude or cross zero. No equivalence margin was pre-specified, so an unresolved comparison is not an equivalence result.

5.6 Evidence Separation and Verification Scope

Confirmation answers and supporting-fact labels were separated from retrieval, prompt construction, and generation until their outputs were fixed. Designated development labels could select configurations; therefore, label-free retrieval does not mean that no development labels were used anywhere in the research process. Post-decision support audits were not used to retune the corresponding completed evaluation.

The scoring and prompt-composition checks use stored predictions and verified reference files. Their scope and artifact traceability are documented in Appendix A; they do not supply a new retrieval or generation experiment.

Setting and method	F1	EM	CR@20	ER@20
HotpotQA / Dense	.42150	.35600	.73700	.87860
HotpotQA / Full	.43628	.36600	.79800	.90818
MuSiQue / Dense	.13595	.10033	.58900	.80553
MuSiQue / Full	.14735	.11033	.65000	.84128
Joint / Dense	.30446	.25467	.66617	.84529
Joint / Full	.31803	.26400	.72217	.87657

Table 3: Compact MiniLM with Qwen. Joint metrics are dataset-equal-weighted. All values are on the 0–1 scale. HotpotQA and MuSiQue support coverage use different annotation granularities. Answer scores reproduce the historical values under dataset-canonical rescoring of fixed predictions.

6 Answer-Quality Results

6.1 Repeated Compact-Backbone Improvements

Table 3 reports the compact results on their original scale. The initial HotpotQA effect is $+0.01478$ with interval $[0.00020, 0.02988]$, and the MuSiQue effect is $+0.01140$ $[0.00450, 0.01835]$. The separate joint component study gives $+0.01357$ $[0.00491, 0.02233]$. Every recorded interval lies above zero, although the initial HotpotQA lower bound is close to it. Together, the three comparisons support a repeated benefit for this compact pipeline under their recorded metrics.

The gain is not uniform over questions. The existing descriptive reconstruction counts 47 positive and 34 negative answer-F1 changes in the initial HotpotQA set, 125/77 in MuSiQue, and 102/73 in the joint set. These counts explain why a positive average is compatible with individual harms. They are answer-change counts, not counts of additional supporting evidence; the latter use a different event definition.

The mean placed-candidate counts are 1.9300, 2.4110, and 2.1628 for these three settings. The joint insertion mean and event counts are query-weighted over 2,500 questions, whereas the joint answer metrics in Table 3 are dataset-equal-weighted. This distinction prevents a larger dataset from implicitly changing the primary effect’s meaning.

6.2 Facet and Protection Comparisons

Compact FULL minus NoFacet is $+0.01336$ $[0.00341, 0.02343]$. The interval lies above zero for the specific centroid-only comparator. It supports an incremental role for facet-conditioned selection in that implementation. It does not isolate the intrinsic value of a hypergraph representation from all coverage or diversity alternatives. The compact NoProtection contrast is $+0.00354$ $[-0.00675, 0.01389]$, which remains unresolved.

The cross-space facet contrast is -0.00743 $[-0.01616, 0.00132]$, and its placement contrast is $+0.01122$ $[0.00129, 0.02104]$. Only the latter interval excludes zero. In native space, the facet pipeline contrast is $+0.00063$ $[-0.00422, 0.00536]$, while the placement contrast is $+0.00337$ $[-0.00129, 0.00803]$; both cross zero. Figure 2 displays all six contrasts to avoid transporting the compact facet result into an unsupported BGE claim.

6.3 Original Completion against a Stronger Baseline

In the joint component set, BGE has answer F1 .35801 compared with .31803 for original FULL. The paired FULL-minus-BGE difference is -0.03998 $[-0.05393, -0.02621]$, with its entire recorded interval below zero. The original MiniLM-based system is therefore worse than BGE under that evaluation. The gain over MiniLM cannot be substituted for an improvement over a stronger retrieval backbone.

The same set gives FULL-minus-BM25 -0.00755 $[-0.02259, 0.00773]$ and FULL-minus-hybrid

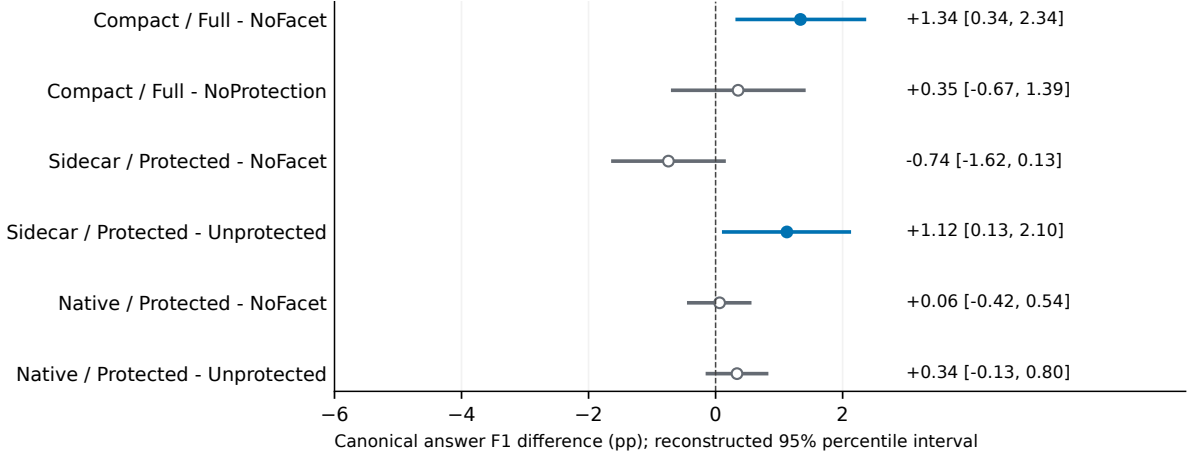


Figure 2: Component and placement effects in percentage points, with recorded 95% intervals, unchanged under canonical rescaling. Open markers cross zero. Native NoFacet changes scoring coefficients in addition to facet eligibility, so it is a pipeline comparison. The cross-space placement contrast has verified identical prompt-visible evidence; this check is specific to that study.

Evaluation	Method	F1	EM	CR@20	ER@20
Joint component	BGE	.35801	.30467	.85400	.93902
	Original Full	.31803	.26400	.72217	.87657
Cross-space	BGE	.34301	.28000	.84967	.93594
	Protected	.34045	.27767	.84350	.93459
Native confirmation	BGE	.34569	.28500	.86167	.94288
	Protected	.34264	.28133	.86233	.94288

Table 4: Absolute strong-backbone results under dataset-canonical scoring, averaged equally across datasets within each evaluation. Different evaluation sets are not pooled. The identical rounded native ER@20 values do not imply identity of evidence sets.

$-0.00320 [-0.01591, 0.00942]$. Both are unresolved. Including these reference points shows that the compact gain is not a general win over all reasonable baselines.

6.4 Cross-Space and Native BGE Extensions

The sidecar tests complementarity while preserving BGE as the primary ranker. Its Protected-minus-BGE effect is $-0.00256 [-0.00998, 0.00458]$ (absolute scores in Table 4). The native reconstruction yields $-0.00305 [-0.00688, 0.00063]$ on its own confirmation set. Neither establishes an increment. Their small negative point estimates and intervals crossing zero do not demonstrate equality or prove that every BGE-compatible structural method would fail.

The native dataset-specific F1 differences are as follows:

$$\begin{aligned} \text{HotpotQA:} & \quad - .002907 \quad [-.008682, .002652], \\ \text{MuSiQue:} & \quad - .003184 \quad [-.008486, .001863]. \end{aligned}$$

The equal-weight EM difference is $-.003667$, with an interval of $[-.007667, .000004]$. These values corroborate the direction of the reported point estimates but do not resolve either dataset-specific comparison. The positive development value used to choose C10 is not confirmation evidence.

The unprotected policies are informative additional references. Sidecar Unprotected minus BGE is $-0.01379 [-0.02440, -0.00339]$; native Unprotected minus BGE is $-0.00641 [-0.01171, -0.00118]$.

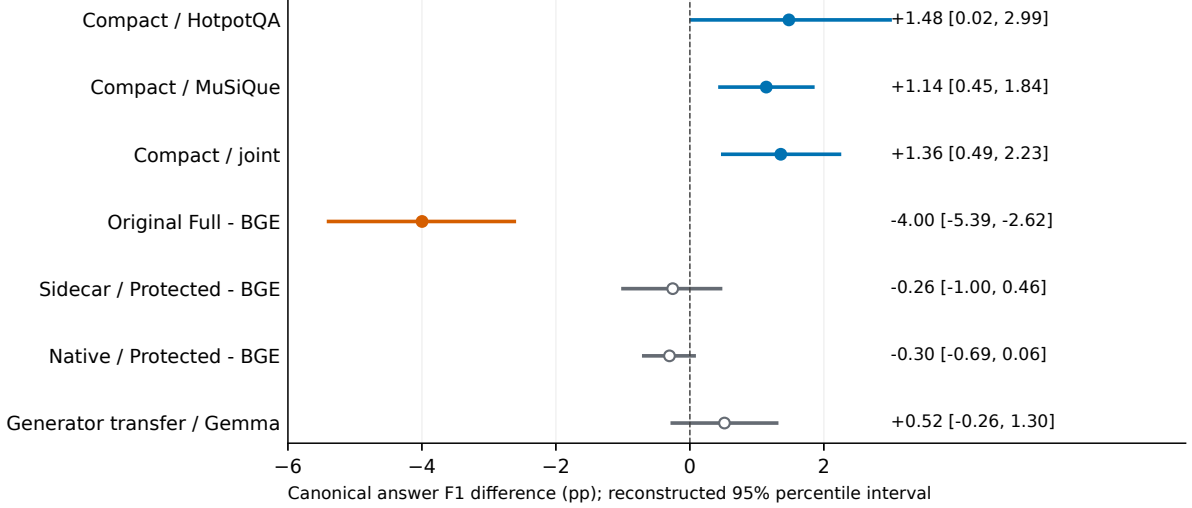


Figure 3: Main paired effects on a common percentage-point display scale. The three compact rows, original-versus-BGE, two BGE extensions, and generator transfer remain separate estimands. Lines are recorded 95% percentile intervals. This is not a fitted relationship between retriever strength and benefit.

Both intervals lie below zero. Protection can reduce harm from an intervention while the protected intervention’s comparison with the original BGE baseline remains unresolved.

6.5 Generator Transfer

The pre-specified Gemma mobile-QAT deployment gives an equal-weight FULL-minus-Dense F1 effect of $+0.00516 [-0.00262, 0.01295]$. This result is unresolved. It is one additional generator configuration, not evidence of universal transfer or an intrinsic architecture ranking. Its reuse of initial rankings preserves the retrieval comparison, but does not add new candidate-selection boundaries. Figure 3 compares the effects and intervals across the separate evaluation settings.

7 Interpreting Selection and Arrangement

7.1 A Fixed-Visible-Evidence Placement Comparison

Write the BGE list as $D = (d_1, \dots, d_K)$ and the insertion sequence as I , with $m = |I| \leq 4$. For $K = 20$, $I \cap D = \emptyset$ and protected prefix ten, the two orders are

$$R_P = [d_1, \dots, d_{10}, I, d_{11}, \dots, d_{20}]_{:20},$$

$$R_U = [I, d_1, \dots, d_{20}]_{:20}.$$

Since $20 - m \geq 10$, both contain exactly $I \cup \{d_1, \dots, d_{20-m}\}$. When fewer than twenty candidates exist, the baseline already covers the universe and eligible outside-list insertions are empty. This argument is specific to these constraints.

Independent reconstruction verifies these premises for every cross-space question: 1,000 HotpotQA and 1,500 MuSiQue cases have identical final members. It also rebuilds each arm’s actual serialized prompt from the bound question, unit titles and texts, template, and local tokenizer. All 5,000 prompt hashes and token counts match their own recorded values. All 2,500 pairs retain the same visible evidence IDs and text multiset, with no dropped or partially truncated unit. Twelve HotpotQA questions have $K < 20$ and no insertions. Ordering changes for 458 HotpotQA and 1,070 MuSiQue questions; the remaining pairs have identical prompts. The primary effect averages all queries.

Sidecar dataset	Min	Median	p95	Max	At cap
HotpotQA	184	923.5	1,156.0	1,907	0/1,000
MuSiQue	504	884.0	1,186.1	2,881	0/1,500

Table 5: Serialized input tokens for Protected and Unprotected. Counts are identical within every paired query, not merely on average. Both arms have zero evidence drops and zero partial truncations. The cap is 4,096 tokens.

The cross-space contrast is therefore an ordering/placement-policy comparison at fixed visible evidence content, including the accompanying position numbers. It does not reveal an internal attention mechanism. The positive Protected-minus-Unprotected estimate coexists with an unresolved Protected-minus-BGE estimate because the latter changes the evidence set.

7.2 Entry Budget and Token Utilization

The twenty-entry bound constrains membership; the 4,096-token cap bounds serialized input length. Table 5 distinguishes them using actual per-query audits, not equality of aggregate averages. The two sidecar orderings have identical token counts for each question, and none reaches the cap. Across all 74,750 main prediction records with prompt audits, including native development, the audits record zero whole-unit drops and zero rank-one partial truncations. Thus, list-level displacement in these runs should not be described as token-capacity exhaustion. This observation does not imply that longer contexts would never truncate or that equal entry budgets imply equal token costs across methods. The record count includes multiple method arms and repeated generator evaluations; it is not an independent question sample size.

7.3 Candidate Placement Is Not Uniform across Systems

The existing Gold-free descriptive trace summaries show different intervention rates. Under dataset-equal weighting, zero candidates are placed for 30.67% of compact queries, 41.43% of sidecar queries, and 80.80% of native queries. Four candidates are placed for 33.95%, 17.80%, and 9.32%, respectively (Figure 4). These percentages describe distinct query sets and distinct candidate rules; they are not the results of a matched intervention-rate experiment.

The sparse native intervention gives context to its small aggregate effect. Many queries preserve the original BGE ranking, while the modified queries still need candidates that improve the answer. A lower intervention rate can reduce both gain and harm. The trace does not establish whether the native gates are too strict, whether available candidates are unhelpful, or whether the generator fails to use them. Relaxing those gates after observing confirmation would be a new selection decision, not further analysis of the same frozen method.

7.4 Supporting-Evidence Addition and Displacement

The post-decision audits compare support identities entering and leaving the final ranking. Sidecar HotpotQA has 21 added and 29 displaced supporting units, for net -8 . Sidecar MuSiQue has 59 added and 50 displaced, for net $+9$. Native HotpotQA has 3 added and 2 displaced, while native MuSiQue has 16 added and 15 displaced, giving net $+1$ in each. Table 6 preserves these totals and their denominators.

These counts do not yield a monotone account of answer utility. The native system adds slightly more supporting evidence than it removes, yet both dataset-specific answer-F1 point estimates are negative. An added supporting unit may be redundant with retained text, and a displaced unit may have a different role from an added one. Context order can also change what the generator uses. No truncation was observed in

Variant	Dataset	Questions	Added	Displaced	Net
Sidecar	HotpotQA	1,000	21	29	−8
Sidecar	MuSiQue	1,500	59	50	+9
Native	HotpotQA	1,000	3	2	+1
Native	MuSiQue	1,500	16	15	+1

Table 6: Existing post-decision supporting-evidence events for Protected relative to BGE. Counts use each dataset’s support identities, so a MuSiQue supporting paragraph is not a HotpotQA supporting sentence. These are descriptive totals, not causal effect estimates or new confirmation tests.

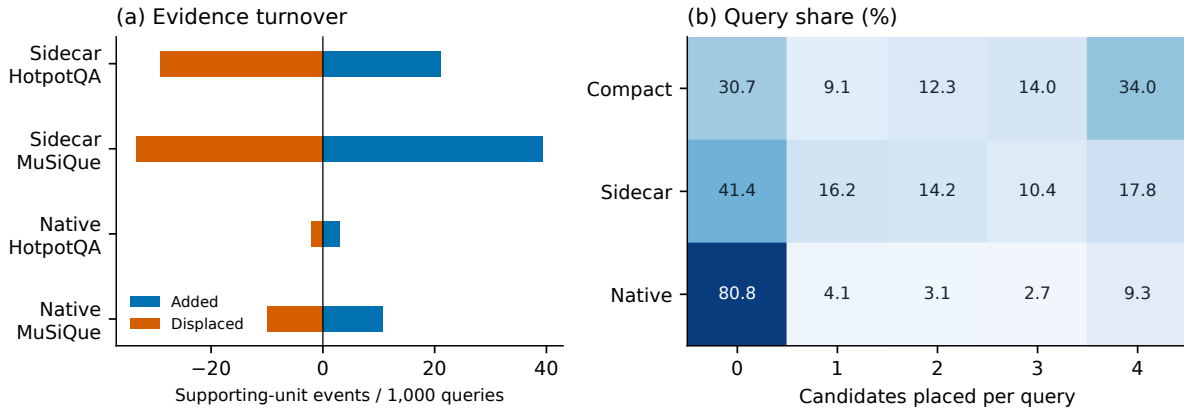


Figure 4: Existing descriptive mechanism summaries. Left: added and displaced annotated evidence per 1,000 questions, with direction represented by sign. Right: dataset-equal-weight query percentages by placed-candidate count. Candidate placement can include within-Top-20 promotion in the compact variant; it is not interchangeable with genuinely new evidence.

the reviewed prompts, and the fixed-content sidecar contrast separately estimates its placement-policy effect. The turnover totals alone still do not identify mediation by any particular added or displaced unit.

The practical implication is to report four objects separately: group-level facet coverage, selected-unit coverage, prompt-visible evidence, and answers. Only the last supplies the present primary endpoint. Treating an improvement in the first as a guarantee about the last would overlook both displacement and generation.

7.5 Representative Success and Failure

The cases in Table 7 use the example closest to the median answer-F1 change within a specified event family, with a stable identity tie-break. The success family requires a positive compact Full-minus-Dense answer change, improved complete recall, and an inserted supporting unit. The failure family requires a negative original-Full-minus-BGE answer change and a displaced supporting unit. This procedure selects illustrations after the outcome; it does not create an unbiased estimate of prevalence.

The success question asks which film by the director of *You and I* concerns the Khmer Rouge. Dense retrieves a description of *The Killing Fields* at rank nine but answers UNKNOWN. Completion adds, at rank twelve, a sentence about *You and I* that identifies Roland Joffé and connects him with *The Killing Fields*. The generated answer changes to the reference film, with F1 moving from zero to one. The added bridge makes the answer derivable from the displayed evidence, although one example cannot isolate a general causal mechanism.

The failure question asks whom the chairman of the Red Cross Aid to Russia Fund was married to. BGE retains a rank-two sentence identifying Clementine Churchill as Winston Churchill’s wife and answers correctly. The original compact Full ranking omits that supporting sentence and produces UNKNOWN; its

Case	Evidence relation	Answer transition
Compact success	Existing Khmer Rouge film description plus an added director–film bridge	Dense: UNKNOWN $\rightarrow$ Full: The Killing Fields; F1 0 $\rightarrow$ 1
Cross-backbone failure	BGE marital relation omitted by original compact Full; unrelated society sentence added	BGE: Sir Winston Churchill $\rightarrow$ original Full: UNKNOWN; F1 1 $\rightarrow$ 0

Table 7: Paraphrased illustrations from the existing representative-case records. The second case is not a protected-placement ablation. Original questions, evidence text, method identities, and selection rules remain in the shared source CSV.

Recorded item	Value	Interpretation
Retrieval reconstruction	8.544 s	Cached native transaction
BGE generation	887.650 s	Per-method aggregate
Protected native generation	884.198 s	Per-method aggregate
Peak GPU memory	3.592 GiB	Recorded hardware
Embedding cache	670,420,228 bytes	Existing cache identity
Main transaction generation calls	10,000	Four arms $\times$ 2,500 queries
Generated in final recorded process	2,000	Resume-process counter
Recorded failed calls	0	Historical telemetry

Table 8: Native confirmation cost record. Transaction-wide and final-process counters are different fields; the table does not imply that 10,000 calls occurred in the final resumed process. No generation was run for manuscript preparation.

added sentence about a different Red Cross society does not repair the link. This is a comparison of different retrieval pipelines, not an instance of the protected sidecar evicting BGE’s protected rank-two unit.

7.6 Computational Cost and What Was Measured

For $n = |U_q|$, embedding dimension d , maximum depth H , and B leaf balls, dense scoring and sorting cost $O(nd + n \log n)$. Recursive assignment costs $O(Hnd)$ because every level processes at most all candidates. Facet scoring and ordering contribute $O(n + Bd + B \log B)$ apart from tokenization; bounded insertion is linear in the traversed candidate list. Working storage is $O(nd + n + B)$. These bounds exclude encoder and generator inference, which have different computational profiles.

The native cost record (Table 8) reports 8.544 seconds for retrieval reconstruction and a 670,420,228-byte embedding cache. Generation totals are 887.650 seconds for BGE, 884.198 for Protected, 886.058 for NoFacet, and 888.282 for Unprotected. Peak GPU memory is 3.592 GiB. These measurements describe a cached local transaction; they are not full index-construction costs, uncached retrieval latency, or evidence of a stable speed advantage from a few seconds of timing difference. The runner accumulates per-call generation seconds already stored in its checkpoint before adding resumed calls. Consequently, transaction generation seconds and the final process wall time have different scopes; their mismatch is not by itself a timing error.

Cache identity and deterministic subsets help interpret execution records, but cache hits are not independent generated-answer reproductions. The historical full or pre-selected subset rerun checks remain documented in their original records. These measurements describe the evaluated implementations and do not benchmark subsequent algorithm revisions.

8 Discussion

8.1 What the Compact Gain Establishes

The repeated compact differences demonstrate that an inference-time completion layer can improve the tested MiniLM system without replacing its encoder. The facet contrast further favors query-conditioned selection over the specific centroid-only alternative. This is a meaningful result about a constrained evidence-selection problem: a flat relevance list can leave recoverable information outside the final context.

The available comparisons do not establish that granular-ball organization outperforms ordinary clustering or direct unit-selection rules. Ordinary grouping, relevance–diversity selection, and unit-level coverage remain plausible explanations for some or all of the increment. The scalar compactness properties do not settle that question, because they hold for arbitrary partitions. The appropriate interpretation is therefore a useful complete pipeline plus limited component evidence, rather than proof that every structural component is irreducible.

8.2 Interpreting the Strong-Backbone Boundary

The BGE results change the practical claim. A user choosing between original compact Full and BGE should not infer superiority from the MiniLM-relative gain: the direct comparison favors BGE. The sidecar and native studies ask whether structure can then complement that stronger reference. Their intervals do not establish the desired increment.

One explanation is that BGE already retrieves much of the available useful evidence, leaving fewer opportunities for completion. Other explanations are poor candidate scoring, overly restrictive gates, insufficiently informative lexical facets, or a mismatch between the selector and generator. The existing comparisons do not identify which explanation dominates. Because both algorithm definitions and query sets vary, a universal law that benefit decreases with retriever strength would exceed the evidence.

The negative original-versus-BGE result and unresolved extension results also have different meanings. The former interval is wholly negative; the latter permit small effects in either direction. Neither should be hidden behind a single qualitative success/failure label.

8.3 Selection and Placement Are Distinct Decisions

Protected placement improves on front insertion in the cross-space study, while its comparison with BGE does not establish an answer-quality gain. Thus, the evidence supports the tested ordering policy relative to front insertion while leaving BGE-relative candidate utility unresolved. The native protection contrast remains unresolved, showing that the placement finding is not automatically portable between representations.

The cross-space study satisfies the fixed-visible-content condition: the retrospective per-question reconstruction verifies identical evidence members and texts under both orderings. It therefore estimates the effect of the tested ordering/placement policy, including the associated position numbers in the prompt. The effect uses all original queries, including those whose order is unchanged. It does not identify an internal attention mechanism. A pure evidence-set intervention would instead hold placement fixed and change the selected units; that is a different comparison. Neither finding establishes that Protected improves upon the original BGE ranking.

8.4 Implications for Future Matched Comparisons

The smallest informative extension would compare completion rules on a fresh compact boundary while matching candidate access, final K , protected prefix, changed positions, genuinely new units, and token accounting. Ordinary clusters should match the actual group count of each candidate configuration; a

Scientific question	Existing evidence	Uncompleted matched comparison
Selection beyond relevance	Repeated compact Full–Dense gains	MMR and actual selected-sentence coverage
Value of grouping	Specified compact NoFacet contrast	Ordinary clusters at actual group count; size-matched random groups
Pure component effect	Native comparison changes several weights	Single coefficient removal at fixed remaining parameters
Ordering vs membership	Fixed-visible-content placement effect	Evidence-set intervention at fixed placement
Quality–cost trade-off	Prompt lengths and local transaction timings	Larger Dense list and shared token-cap comparisons
Strong-score increment	Original Full negative; BGE extensions unresolved	Common strong candidate pool and base scores

Table 9: What the completed evidence answers and what remains unmeasured. Uncompleted comparisons are not negative results. None is counted as an additional contribution or confirmation experiment.

random grouping can additionally match group-size distribution. Pure component ablations should zero the designated coefficient while leaving the others unchanged. Independently tuned pipelines answer a separate question and should be reported separately.

Such a study would need a prospective development/confirmation split, a minimum meaningful effect, a finite tuning budget, and stopping rules. Formal uncertainty should follow a justified sampling model and null hypothesis. These are requirements for a future experiment, not additional results of this article. Corpus-scale retrieval and external structured methods would additionally require aligned corpora, readers, and budgets.

The comparison hierarchy matters. An exact reproduction preserves the external method’s extraction, corpus and reader. An adapted control may share our candidate pool and reader, but must identify every change to that method. A conceptual comparison, such as Table 1, identifies algorithmic differences without providing a performance ranking. Merely sharing an encoder does not align different corpora or context budgets. Table 9 connects these requirements to the questions left open by the completed evidence.

9 Limitations

The evidence is restricted to closed candidates in two English multi-hop QA benchmarks. We do not evaluate corpus reachability, ANN recall, index storage, full-wiki answer quality, multilingual transfer, or unanswerable questions. MuSiQue’s paragraph-level support annotations cannot be reinterpreted as sentence-level reasoning-chain labels. Query terms also need not contain the bridge entity required for a multi-hop answer.

Only one compact and one strong backbone are evaluated, with one principal generator and one additional deployment configuration. The results cannot establish broad generator robustness or a continuous relation between retriever capacity and completion utility. Training-release questions may have appeared in generator pretraining; project-held-out selection does not eliminate that possibility.

Several mechanism claims remain unidentified. The compact radius split trigger is redundant, the selector is static, group facets may exceed selected-unit facets, and the native NoFacet comparator changes multiple weights. There is no completed matched flat-unit, ordinary-clustering, MMR, or sentence-coverage experiment. The mathematical coverage bound belongs to a distinct surrogate and supplies no guarantee for historical answer F1.

The measured scoring discrepancy is confined to one development question and leaves the reported method differences unchanged. Historical tail-probability calibration and source-document dependence limit inferential claims. We do not silently replace the historical statistics with a new test. The confidence

intervals are not equivalence bounds or uncertainty over model-development decisions. Mechanism audits and qualitative examples are post-decision descriptions, not new causal tests.

10 Conclusion

HyperGranular-RAG frames structured evidence completion as a bounded intervention over a dense retrieval ranking, linking evidence selection with evidence arrangement. The compact implementation repeatedly improves MiniLM answer F1 in the evaluated closed-candidate settings, and facet-conditioned selection exceeds its specified compact centroid-only control. The original system nevertheless falls below BGE, and the tested cross-space and native extensions do not establish an increment over BGE. At fixed visible evidence, protected placement improves on front insertion in the cross-space setting, while its answer-quality comparison with BGE remains unresolved.

The analysis explains why these outcomes are compatible. Group compactness, lexical coverage, candidate addition, prompt-visible evidence, and answer quality are different quantities. A method can improve one without improving the next. Precise algorithm versions and matched comparisons are therefore essential to attributing a gain. The existing evidence supports completion for the tested compact system and motivates a narrower, testable question about which candidate interventions can complement strong retrieval.

Ethics, Availability, and Research Responsibility

The study uses existing benchmark releases and documented model artifacts; no new participant study is introduced. Positive, negative, and unresolved results are retained. Third-party data and weights remain governed by their original terms. The research repository contains implementations, frozen derived records, and source data for the reported figures. An anonymous artifact distribution and final reuse-license statement remain matters for the human authors before submission.

AI tools assisted software work, analysis preparation, and manuscript writing. They are not authors and do not bear responsibility for the work. Human authors must confirm authorship, affiliations, funding, conflicts, and the final venue-specific disclosure. No unconfirmed declaration is inferred here.

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A Historical Evidence Mapping

The paper’s scientific comparisons correspond to the following internal records: initial compact HotpotQA and MuSiQue are Stage4E and Stage4F; generator transfer is Stage4G; joint components are Stage4H; the cross-space sidecar is Stage4I; and native confirmation is Stage5A. These identifiers are provenance labels, not the scientific organization of the argument. The shared manuscript ledger lists the exact aggregate paths, effect values, and historical-method interpretation.

The baseline repository commit for the preceding audit was:

8a8e7cef8551c14bcea99e6a41a2b5a83f65456f

The old PDF and source are retained separately from this manuscript version. Some aggregate files retain a pending-verification status because verification was written to a separate final artifact; the aggregation file is not rewritten to erase that transaction history.

Later Stage6 development artifacts, its partially prepared official pipeline, and the repaired v2 synthetic prototypes are not included in the empirical tables. Their mathematical counterexamples help delimit possible claims, but their synthetic behavior is not evidence about held-out answer quality. No restricted reservation or Stage3B data is used in this manuscript revision.

Subsequent bounded proposal. A follow-up proposal retained simple matched selectors, larger Dense lists, shared token caps, and one shared strong-score setting. Execution did not begin: the prior exposure history of the available confirmation set could not be verified, and no substitute set was sought. Resource ceilings were agreed, but no development choice or confirmation was performed. This is an eligibility limit, not evidence of method failure. The proposal’s dynamic coverage and bridge terms are not used to reinterpret the evaluated static selector or to claim a new completed algorithmic contribution.

Scorer sources and reconstruction. The first HotpotQA and generator-transfer scorers apply the official special-answer exception: an unequal normalized prediction involving yes/no/noanswer receives zero F1. The initial MuSiQue scorer uses its dataset-specific normalization and aliases. The component, sidecar, and native scorers instead use alias-max normalized token overlap without the HotpotQA exception. Such overlap can assign partial credit to a longer affirmative phrase, where the official HotpotQA rule assigns zero.

Reconciliation covers 74,750 prediction records, not independent questions. All 53,500 confirmation and generator-transfer records retain their EM/F1. On native development, one HotpotQA question produces the same longer affirmative answer for all 17 methods; each F1 falls from .20 to zero. Each HotpotQA method mean falls by .0004, but all paired differences and the reported C10 development difference remain unchanged. No configuration selection was rerun. The other 21,233 development predictions are unchanged. Prediction strings are not newly cleaned or rewritten.

HotpotQA uses the official `hotpot_evaluate_v1.py` at revision:

3635853403a8735609ee997664e1528f4480762a.

MuSiQue uses `metrics/answer.py` at revision:

922ac98f19a201998dbdae6d7f2887a5258dbdeb. The revision package records their source URLs and SHA-256 values and executes their reviewed pure answer functions. Gold files match the historical frozen identities before parsing. Per-query legacy values, all method means, original confirmation percentile intervals, and generator-interaction intervals are reconstructed independently. Legacy and canonical scores use exactly the same resampling draws. The 17 affected development predictions all belong to the same question; their common F1 correction cancels in every within-query method contrast. Raw reference maps are not distributed with the revision package.

The independent check addresses numerical scoring and prompt composition. It neither certifies unrelated later verifier repairs nor supplies a null-calibration proof for archived bootstrap-tail/Holm decisions.

B Terminology and Reporting Units

Term	Meaning used in this article
Evidence completion	Bounded change to an existing evidence list; no promise of complete reasoning support.
Granular ball	Local embedding group with a computed center and radius; historical compact split eligibility is not independently radius-triggered.
Facet	Fixed-tokenizer query term occurring in a group’s title/sentence union, not a verified semantic subquestion.
Facet hyperedge	Seed-to-expansion group relation based on seed-relative facet novelty and frozen gates.
Protected prefix	Initial dense entries preserved in order by the placement rule.
Placed candidate	Candidate accepted into the insertion segment; may be a promotion from dense ranks 11–20 in the compact method.
Genuinely new unit	Unit in the completed list but absent from the baseline Top-20.
Evidence coverage	Coverage of annotated supporting identities at the dataset’s sentence or paragraph granularity.
Answer utility	Observed change in answer EM/F1, not a score used by the confirmation retriever.
Unresolved effect	Recorded interval crosses zero; not evidence of equivalence.
Legacy scoring	Frozen evaluator used to reconstruct historical values; the component, sidecar, and native versions lack HotpotQA’s special-answer rule.
Canonical answer scoring	Dataset-specific official answer functions applied retrospectively to unchanged predictions and verified reference files.

C Counterexamples for Unevaluated Extensions

A two-group activation objective illustrates a limitation of possible extensions. For $e = \{a, b\}$, let $H(S) = \mathbf{1}[|S \cap e| \geq 2]$. Adding b to the empty set has gain zero, while adding it to $\{a\}$ has gain one, violating diminishing returns. Complementarity can be desirable without being submodular.

Degree normalization presents a separate issue. Take hyperedges $\{a, b\}$ and $\{b, c\}$, and divide newly activated edges by the incoming group’s degree. The path $a \rightarrow b$ accumulates $1/2$, whereas $b \rightarrow a$ accumulates 1 for the same final set $\{a, b\}$. Such scores cannot be exact marginals of an order-independent set potential. These are not the rules of historical compact q25. They supply no evidence about its measured gains and no new empirical contribution. A bounded lookahead alternative would require its own objective and cost evaluation.

D Reproduction Details

Retrieval and units. The compact encoder is all-MiniLM-L6-v2, with attention-mask mean pooling, L2 normalization, and a 192-token encoder limit. BGE uses bge-large-en-v1.5, CLS pooling, L2 normalization, and a 512-token encoder limit. Its query instruction is “Represent this sentence for searching relevant passages:” followed by the question. Document inputs concatenate the title and sentence. HotpotQA preserves the supplied sentence boundaries; MuSiQue splits whitespace-normalized paragraphs with the frozen punctuation-and-capitalization rule. Neither splitter is changed in this review. The reproducibility configuration pins the encoder revisions and unit IDs.

Lexical baselines. BM25 uses lowercase alphanumeric tokens, $k_1 = 1.5$, and $b = 0.75$ within each query’s candidate set. Its inverse document frequency is $\log(1 + (N - df + 0.5)/(df + 0.5))$. The hybrid baseline averages query-local min–max-normalized dense and BM25 scores with equal weights; a constant score vector maps to zero. Unit ID breaks score ties. These are local closed-candidate baselines, not corpus-scale retrieval systems.

Generation and budgets. The principal reader is Qwen2.5-1.5B-Instruct in FP16, using greedy decoding, one beam, batch size one, and at most 32 generated tokens. The full serialized input, including the chat template, is capped at 4,096 tokens. The system instruction is: “Answer the question using only the provided evidence. Return only the shortest final answer. If the evidence is insufficient, return UNKNOWN.” Evidence is rendered in rank order as numbered title–sentence entries, followed by the question and “Final answer:”. The Qwen runner sets CPU/CUDA seeds to zero, uses one CPU thread, and enables deterministic algorithms. The renderer admits whole units in order until the cap; only an overlong first unit permits partial truncation. The compact policy protects ten dense positions and places at most four candidates under a final budget $K = \min(20, |U_q|)$. Its frozen q25 floor is 0.1957079917192459. BGE-native has its own recorded eligibility policy; it is not retrospectively identified with compact static-q25.

Scoring and interpretation. The reanalysis fixes the original predictions and reference channel. It executes the pure answer-metric functions from the pinned official HotpotQA and MuSiQue implementations, with maximum-over-alias scoring only where the dataset permits aliases. HotpotQA assigns zero token F1 to mismatched yes/no/noanswer responses, and zero F1 when both normalized strings are empty. MuSiQue’s official token F1 assigns one to the latter case and does not inherit HotpotQA’s categorical exception. The archived per-query scores, means, and paired percentile intervals are first independently reconstructed. Canonical scores then reuse the same 10,000 bootstrap draws. This isolates scorer migration from statistical redesign: no new confirmatory test, model selection, or independent confirmation is claimed. Supporting-evidence coverage retains the dataset’s original sentence or paragraph granularity and is not a complete-chain recovery metric.